Aalto-yliopisto
Aalto-universitetet
Aalto University

Bachelor's Programme in Economics

Evaluating "Fair Chance" Employment

Revisiting Ban the Box's Effect on Minority Employment with Extended ACS Data

---

Nooa von Schöneman

Bachelor's thesis
2026

Copyright ©2026 Nooa von Schöneman

This work is licensed under a Creative Commons “Attribution-NonCommercial-ShareAlike 4.0 International” (CC BY-NC-SA 4.0) license.

https://creativecommons.org/licenses/by-nc-sa/4.0/

Aalto-yliopisto
Aalto-universitetet
Aalto University

Author Nooa von Schöneman

Title of thesis Evaluating “Fair Chance” Employment: Revisiting Ban the Box's Effect on Minority Employment with Extended ACS Data

Programme Bachelor’s Programme in Economics

Major Economics

Thesis supervisor Prof. Mitri Kitti

Thesis advisor(s) Prof. Mitri Kitti

Date 19.08.2026 **Number of pages** 33 **Language** English

Abstract

Numerous states and sub-regions in the U.S. have adopted policies that aim to postpone when and if employers can access the criminal histories of applicants. These “Ban the Box” laws are designed to help individuals with criminal records get into the labor force but have been met with concerns that removing the ability to provide a signal of a clean record could harm minority candidates if employers began substituting the missing information with observable proxies like race. Prior research done on the subject contains mixed results, with some finding evidence of statistical discrimination and others not.

This thesis expands on the work of Burton and Wasser (2025) by examining the employment outcomes of young, lowly educated minority men following the implementation of Ban the Box laws using American Community Survey data for the years 2004-2019. The results show no robust employment effects of Ban the Box laws for any of the demographics, complementing the findings of Burton and Wasser (2025).

Keywords ban the box, statistical discrimination, fair-chance hiring, employment, difference-in-differences, criminal record

Aalto-yliopisto
Aalto-universitetet
Aalto University

Tekijä Nooa von Schöneman

Työn nimi ”Fair Chance” -työllisyyden arviointi: Ban the Box -lakien vaikutus vähemmistöjen työllisyyteen laajennetulla ACS-aineistolla

Koulutusohjelma Taloustieteen kandidaattiohjelma

Pääaine Taloustiede

Vastuuopettaja/valvoja Prof. Mitri Kitti

Työn ohjaaja(t) Prof. Mitri Kitti

Päivämäärä 19.08.2026 **Sivumäärä** 33 **Kieli** Englanti

Tiivistelmä

Useat Yhdysvaltain osavaltiot ja alueelliset toimijat ovat ottaneet käyttöön lakeja, joiden tarkoituksena on lykätä sitä, milloin ja miten työnantajat voivat tarkastaa työnhakijoiden rikostaustan. Nämä niin sanotut ”Ban the Box” -lait on suunniteltu auttamaan rikosrekiserin omaavia henkilöitä pääsemään työmarkkinoille, mutta ne ovat herättäneet huolta siitä, että puhtaan rikosrekisterin signaloinnin poistaminen saattaisi vahingoittaa vähemmistöhakijoita, mikäli työnantajat alkaisivat korvata puuttuvan tiedon havaittavilla piirteillä, kuten ihonvärillä. Aiheesta tehty aiempi tutkimus on tuottanut ristiriitaisia tuloksia: osa tutkimuksista on löytänyt näyttöä tilastollisesta syrjinnästä, toiset taas eivät.

Tämä opinnäytetyö laajentaa Burtonin ja Wasserin (2025) tutkimusta tarkastelemalla nuorten, matalasti koulutettujen vähemmistöihin kuuluvien miesten työllisyyskehitystä Ban the Box -lakien käyttöönoton jälkeen käyttäen American Community Survey -aineistoa vuosilta 2004–2019. Tulokset eivät osoita Ban the Box -lakien vaikuttavan merkittävästi minkään väestöryhmän työllisyyteen, mikä tukee Burtonin ja Wasserin (2025) havaintoja.

Avainsanat ban the box, tilastollinen syrjintä, reilun mahdollisuuden palkkaus, työllisyys, difference-in-differences, rikosrekisteri

Table of contents

Preface and acknowledgements 6
1 Introduction 7
2 Literature review 10
2.1 A brief review of Ban the Box studies 11
2.2 The macro studies 12
3 Data 15
4 Empirical strategy 17
5 Results 21
6 Discussion 25
7 Conclusion 28
Artificial intelligence disclaimer & references 29
Use of generative artificial intelligence 29
References 29
Appendix 32

Preface and acknowledgements

The fact that I am currently writing this thesis with my biggest worry being the summer heat is the sum of more things aligning favourably for me than I could possibly realize. It would not be possible without the lifelong support of my close ones, and the countless living or now dead beings who have or had influenced the world we currently live in.

150 years ago, medicine was still in its infancy, and a single bad crop could have doomed a family's fate. 150 years from now the Earth may be much harsher of an environment to live in due to climate change. But on this brief window of time we find ourselves placed in, things are going pretty well.

I am grateful.

Nooa von Schöneman
20.7.2026

1 Introduction

Around one third of all adults in the United States have a criminal record (Avery 2024). Until recently, it was commonplace in the United States to ask about past criminal records in job applications (Connerley et al. 2001). Some employers used this question as a screening tool, eliminating everyone with a record regardless of their possible qualifications (e.g., Pager 2003; Uggen et al. 2014). As such, historically the presence of a criminal record had made it harder to obtain a job, all else being equal.

This issue is especially pressing for ex-convicts, who are often in dire need of economic stability. Among ex-convicts, employment is linked with reduced recidivism. For instance, Berg and Huebner (2011) calculated that 600 days post-release, employed ex-convicts were 23.7% less likely to have been rearrested compared to unemployed counterparts. Consequently, helping reintegrate ex-convicts into the labor force is desirable from both macroeconomic and public safety perspectives.

"Ban the Box" (BTB) is a nationwide employment movement, which aims to postpone any background checks and questions regarding past convictions in the initial hiring process. The theory is that after getting an interview and making a personal connection, rejection based on a background check becomes harder, which helps ex-offenders get a foot in the door in job seeking (Agan and Starr 2018).

The campaign began in Hawaii in the late 1990s, from which it spread to other U.S. states in the coming years (Harless 2013). Hawaii was the first state to pass BTB legislation back in 1998, and after 2004 other cities, counties and eventually states started following suit exponentially (Avery and Lu, 2021). By October 2021 a total of 37 states and over 150 cities and counties had adopted different versions of Ban the Box laws, covering over 80% of the U.S. population (Avery and Lu 2021).

Different jurisdictions have implemented different versions of BTB. There is variation in both what the laws limit, and to which employers they apply to. The laxer laws remove questions about convictions from job applications, but stronger ones also limit the accessibility of conducting background checks. Some laws only apply to the public sector, others extend to include private firms that do contract work for the public sector, and the strictest ones extend to include the entire private sector (Avery and Lu 2021).

The exponential implementation of BTB signals strong public support, but removal of information in the job application process can also trigger unforeseen consequences. Employers with a preference for applicants without records may begin practicing statistical discrimination by substituting the missing information with noisy observable proxies, such as race, gender, and education. This could result in employers favoring certain demographics over others as they get labeled as riskier hires, a phenomenon called statistical discrimination (Agan and Starr 2018; Doleac and Hansen 2020).

This has raised concern that BTB could harm minority candidates with clean records as they lose their ability to signal their clean record in the job seeking process (Agan and Starr 2018; Doleac and Hansen 2020). There is prior research that has found evidence of statistical discrimination in reaction to other similar policies that removed signals, such as credit checks or drug tests (Bartik and Nelson 2025; Wozniak 2015). However, BTB laws tend to be hard to enforce, and crime-averse employers are often still able to circumvent them if they wish to do so (Herring and Smith 2022). As such, it would be more accurate to view criminal history as a quasi-public signal as opposed to a private one. Thus, the effect of its removal may be smaller than in the removal of private signals, which are not circumventable.

The research done on the effects of BTB remains sparse and yields mixed results. Of the nine papers reviewed in chapter 2, two find positive employment effects, four find no effect, and three find a negative effect. Notably for this thesis, Doleac and Hansen (2020) conducted macro level

analysis from 2004-2014 using Current Population Survey (CPS) data and found evidence of BTB harming the employment outcomes of young, lowly educated black and Hispanic men. However, Burton and Wasser (2025) found that these effects were not robust across an alternative American Community Survey (ACS) dataset and Kaestner and Wang (2024) found that the CPS results do not hold up when the sample period is extended until 2019.

This thesis contributes to the literature in three ways. First, I extend the ACS analysis of Burton and Wasser (2025) from 2004-2014 to 2004-2019, which requires coding all BTB adoptions from 2014 to 2019 and widening the event-study window accordingly. Second, I subject the extended sample to a fuller battery of validity checks than the original replication. Third, I benchmark the resulting ACS estimates directly against the CPS estimates of Kaestner and Wang (2024) for the same extended period.

The thesis mainly complements the findings of Burton and Wasser (2025), and Kaestner and Wang (2024) in that BTB laws do not seem to have significant macro level effects on the employment of young, lowly educated men in the medium to long run. The main takeaway is that prior evidence of significant effects on employment found in studies like Doleac and Hansen (2020), and Shoag and Veuger (2021) do not seem to hold up when using a dataset with more statistical power across larger sample periods.

The structure of the thesis includes the following. Chapter 2 reviews previous literature that studies the effects of BTB. The chapter is divided into two sections with the first section briefly reviewing a broader set of BTB literature and the second section reviewing the macro-studies that are relevant for the following chapters in greater detail.

Chapter 3 pivots into original empirical work and explains the data used for my analysis, while chapter 4 explains the methodology. Moving on, chapter 5 goes through the results and chapter 6 contains discussion on how these results fit into the big picture. Finally, chapter 7 concludes the thesis with a brief recap.

2 Literature review

Despite the wide implementation of BTB laws, covering over 80% of U.S. population as of 2021 (Avery and Lu 2021), the research done on the effects of BTB laws remains relatively sparse. A handful of studies published over the past decade address somewhat distinct research questions, yielding mixed results. However, none of the most recent studies have identified statistically significant impacts, which has arguably shifted the consensus toward the view that the overall effects of BTB may be smaller than previously thought.

Generally, researchers studying BTB policies must choose between tracking individuals with criminal records directly or using broader demographic proxies. Both approaches come with their own trade-offs. Datasets that have individual criminal histories tend to either be relatively small or local but have the advantage of directly tracking the target group BTB laws are designed to help. On the other hand, changing the research target to broader demographic proxies allows researchers to utilize much larger, nationwide datasets. However, this makes it impossible to study the effects of BTB on individuals with criminal records directly as they get aggregated with broader populations. This causes the true policy effect sizes to attenuate.

The most relevant studies to this thesis are Doleac and Hansen (2020), Kaestner and Wang (2024), and Burton and Wasser (2025). These three studies evaluate the effects of BTB on the macro-level using large, nationwide datasets. Section 2.1 provides a brief review on a broader number of studies, while Section 2.2 reviews these three core papers in greater detail.

2.1 A brief review of Ban the Box studies

I was able to find a total of nine relevant studies that examine how BTB laws have affected different groups. I will begin with studies that use datasets with criminal histories, and pivot to studies with proxies afterwards.

Jackson and Zhao (2017) studied how Massachusetts' BTB law affected individuals with criminal records and found a 2.6% decline in their employment rate following the implementation of BTB in 2010. Another similar case study was Rose (2021) which examined the effects of Seattle's BTB law on record holders. Rose found no statistically significant findings on income nor employment of ex-offenders, even when broken by race.

Craigie (2020) used a nationwide sample with individual criminal records and around 11,000 person-year observations from 2005 to 2015. The study found a 3.8% increase in the probability of public employment among record holders and no evidence of statistical discrimination. Finally, Weiss et al. (2025) studied the effects of BTB on a public health centre in Texas that kept conducting background checks after a conditional job offer had been given. With a sample size of around 23,000 applications, the study found no effect from BTB on individuals with records receiving a conditional job offer.

Moving on from directly tracking individuals with criminal records, several studies have also examined the effect of BTB laws through proxies on samples of people with relatively high rates of criminal records. Shoag and Veuger (2021) studied the impact BTB had on employment outcomes in neighbourhoods with high crime rates. They reported a 3.5% employment increase post BTB, mostly driven by public sector jobs.

The remaining four studies all studied the impact of BTB on young, lowly educated men of different races. Agan and Starr (2018) conducted a field experiment where they studied applicant call-back rates in entry-level online job postings in New York City and New Jersey. They found that pre BTB, having no record was associated with a 63% higher callback rate when compared to record holders, but also that the black-white callback gap grew from 7% to 43% following the implementation of BTB.

The remaining three studies Doleac and Hansen (2020), Kaestner and Wang (2024) and Burton and Wasser (2025) all used large nationwide datasets to study the impact of BTB. Doleac and Hansen (2020) found a negative effect on the employment of young, lowly educated black and Hispanic men for the years 2004-2014. However, Burton and Wasser (2025) found that this result is not robust across datasets and Kaestner and Wang (2024) found that the results of Doleac and Hansen (2020) do not hold when the sample is extended until 2019.

To summarize, of the nine studies reviewed, two find a positive effect (Craigie 2020; Shoag and Veuger 2021), four find no effect (Rose 2021; Kaestner and Wang 2024; Burton and Wasser 2025; Weiss et al. 2025), and three find a negative effect (Jackson and Zhao 2017; Agan and Starr 2018; Doleac and Hansen 2020).

The different outcomes can be partly explained by the different research questions, but even then, some studies are in direct tension with each other. Several papers have hypothesised that the outcomes of BTB could differ in different sectors and areas with different labor conditions and demographics. Due to this, local micro studies may not be applicable when trying to form a picture of BTB on the macro level. In the next section I will examine the macro studies in greater detail and establish a research gap in their analysis.

2.2 The macro studies

There are two major datasets that were utilized in the macro studies by Doleac and Hansen (2020), Kaestner and Wang (2024), and Burton and Wasser (2025). These are the Current Population Survey (CPS) and American Community Survey (ACS). Both surveys are conducted by the United States Census Bureau. The two datasets contain comparable variables when it comes to employment and demographics and their difference mostly lies in size and frequency. The CPS operates at a monthly frequency with a sample size of around 60,000 households per month. Conversely, ACS is

administered annually but has much more statistical power at around 3.5 million household observations per year.

Doleac and Hansen (2020) based their research on CPS data and used ACS for robustness checking the CPS results. They focus on the employment outcomes of 25–34-year-old men without a college degree, broken by race. The study was built on Metropolitan statistical areas (MSAs) across the entire United States for the years 2004-2014.

Doleac and Hansen (2020) found that for black men in their treatment group, employment probability decreased by 3.4 percentage points post BTB. When the regression was replicated with ACS data, they still saw a 1.3 percentage point decrease that was significant at the 10% level. The paper concluded that the results are robust, with the lower results of ACS caused by attenuation from the annual frequency.

However, Burton and Wasser (2025) contained heavy critique of the Doleac and Hansen (2020) study. They found numerous coding errors in how Doleac and Hansen (2020) had coded BTB laws for various MSAs, both for CPS and ACS. With the corrected code, the results from CPS remained similar, but ACS data failed to reject the null hypothesis of no effect of BTB on employment. In other words, Burton and Wasser (2025) proved that the results of Doleac and Hansen (2020) were not robust across the two datasets.

Furthermore, Burton and Wasser (2025) questioned the validity of the results from CPS analysis based on its low sample size. They noted that CPS is not meant to be representative below state level, while ACS with its bigger sample size is representative of geographies with a population of over 65,000. Burton and Wasser (2025) demonstrated that the CPS analysis of Doleac and Hansen (2020) was conducted with very noisy data. With CPS, the average number of black men sampled in a given MSA-month was just 3.5, and two MSAs measured instances where the black employment rate was at abnormally high 100%. Meanwhile, the ACS samples an average of 28.8 black men in each MSA-year for the same sample period.

Finally, Burton and Wasser (2025) questioned the necessity of monthly frequency for BTB analysis. Despite the attenuation, they were still able to

identify statistically significant effects with the CPS sample when using an annual as opposed to monthly frequency and defining treatment as the first full year when BTB has been in effect.

Furthermore, Kaestner and Wang (2024) extended the CPS analysis of Doleac and Hansen (2020) to include the years 2015-2019, while also using corrected code. They found that the negative employment outcomes for black and Hispanic men from BTB laws that were found in 2004-2014 do not hold up for the extended 2004-2019 period. Kaestner and Wang (2024, 5) concluded that the negative effect found in the earlier period "is spurious and likely a reflection of low statistical power".

To conclude, the two studies of Burton and Wasser (2025) and Kaestner and Wang (2024) do not support the theory proposed in Doleac and Hansen (2020) that statistical discrimination is happening on a macro scale following BTB. While CPS does give statistically significant results for the 2004-2014 period, they do not hold up in the extended 2004-2019 sample, nor when using ACS data for the same 2004-2014 period.

However, to the best of my knowledge nobody has synthesized these two studies and conducted BTB analysis with ACS data beyond the 2004-2014 period. If the ACS data for an extended 2004-2019 period was to also fail to show significant results, it would give more credibility to the hypothesis that the results in Doleac and Hansen (2020) are spurious. During the following sections I will reproduce the ACS results of Burton and Wasser (2025) and extend their research to cover the larger timeframe of 2004-2019 that Kaestner and Wang (2024) use in their CPS analysis.

3 Data

I am basing my research on microdata from the American Community Survey (ACS). ACS is an annual survey of approximately 3.5 million U.S. households conducted by the U.S. Census Bureau. The survey began in 2004 and is still ongoing. I am limiting my research to the 2004-2019 period due to concerns that the COVID-19 pandemic would not be comparable with the earlier years.

As noted, Doleac and Hansen (2020) and Burton and Wasser (2025) used ACS to study BTB for the 2004-2014 period. However, as Table 1 shows, many regions implemented BTB after 2014 and this extended sample should provide a more comprehensive picture of BTB's impact on different demographics.

Mirroring Doleac and Hansen (2020) and Burton and Wasser (2025), I am limiting the sample to black, Hispanic, and white 25–34-year-old men who are U.S. citizens and do not possess a bachelor's degree. The extended sample contains an average of 45.0 black, 46.2 Hispanic, and 187.2 white men per MSA-year cell.

Also mirrored from the studies of Doleac and Hansen (2020) and Burton and Wasser (2025) is the definition of treatment. Treatment is coded at the metropolitan statistical area (MSA) level. An MSA is considered treated if any part of the MSA has implemented BTB as of January 15th that year. This reflects the design choice common to both papers of treating the MSA as the relevant local labor market rather than the state or municipality.

Table 1: Share of sample under treatment (2004-2019 ACS data)

Year	Share under treatment	N
2004	0.005	34,125
2005	0.010	80,586
2006	0.010	87,421
2007	0.025	86,761
2008	0.062	87,454
2009	0.085	89,157
2010	0.110	91,886
2011	0.247	95,847
2012	0.318	94,172
2013	0.385	95,188
2014	0.440	94,603
2015	0.489	94,854
2016	0.629	95,272
2017	0.713	96,492
2018	0.789	97,954
2019	0.821	97,110

As outlined in section 2.2, the biggest advantage of ACS is its sheer size, which makes it representative of geographies with a population of 65,000 and above. Since the analysis is conducted at the MSA level, ACS is quite possibly the only way to conduct meaningful macro analysis of the impact of BTB at this granularity. However, ACS does come with three notable trade-offs.

Firstly, ACS is conducted on an annual basis, whereas CPS has monthly frequency. This creates concerns that ACS results get attenuated when a policy is implemented in the middle of the year. Secondly, the ACS changed the wording of survey questions in 2008, which creates concerns that pre and post 2008 answers might not be directly comparable. Finally, ACS (like CPS) does not have individual crime data, which means that the datasets are unfit to study what impact BTB has had on ex-offenders directly.

Despite the trade-offs, Burton and Wasser (2025) argued that ACS likely remains a better source to study BTB than CPS. For the first trade-off, Burton and Wasser (2025, 5) found strong evidence that "monthly treatment timing is not necessary for recovering treatment effects." The second trade-off can be addressed by running regressions for only post 2008 results on top of the full dataset. Table 1 shows that in 2007 BTB laws were still very sparse and only 2.5% of the sample were covered by them, so the restriction does not take away much information regarding the treatment. Unless otherwise specified, the tables and figures are made using the 2008-2019 specification.

For the final trade-off there is nothing that can be done and I must simply accept the use of proxies.

4 Empirical strategy

I am basing my research on the replication package provided by Burton and Wasser (2025). Mirroring their design, the analysis is conducted via the following two-way-fixed-effects difference-in-differences (DID) model:

$$Y_{itmr} = \beta_1 BTB_{mt} \times Black_i + \beta_2 BTB_{mt} \times Hispanic_i + \beta_3 BTB_{mt} \times White_i + \theta X_{it} + \gamma_{tr} + \delta_m + \lambda_m \times t + \varepsilon_{itmr} \quad (1)$$

In the model Y_{itmr} is a binary indicator for employment status of individual i at time t in MSA m within region r. BTB_{mt} is the treatment variable, equal to 1 if MSA m is covered by a BTB policy at time t, and 0 otherwise. The BTB term interacts with black, Hispanic, and white racial groups to measure the estimated change in employment probability (β) caused by BTB for that racial group. θX_{it} captures the effects of individual-level covariates for age, education, and current school enrollment. The term γ_{tr} measures time-by-region fixed effects, δ_m MSA fixed effects and $\lambda_m \times t$ the MSA-specific linear time trends. Finally, $\varepsilon_{itmr}$is the error term.

The validity of a difference-in-differences research design depends on several key assumptions. The most important one is the parallel trends assumption, meaning that for DID to be valid, the treatment areas' employment outcomes would have to follow the same trend as nearby control areas that did not pass the law in the absence of BTB.

The second assumption is the exogeneity of policy adoption, which requires the adoption and timing of BTB laws to be exogenous to other local changes that might affect employment. For instance, if a city was to pass BTB specifically because its economy was suddenly booming, the estimated BTB effect might be capturing that factor.

The final assumption is the Stable Unit Treatment Value Assumption (SUTVA), which assumes that there are no spillover effects between treated and untreated areas, meaning that the laws should not cause any migration in or out of the treatment areas. I address all three assumptions in turn by

combining my own diagnosis with evidence reported in Doleac and Hansen (2020) and Burton and Wasser (2025).

Like the two outlined studies, I test parallel trends using event-study plots for my extended 2008-2019 ACS sample, which are attached as Figures 1, 2 and 3 below. The figures plot the coefficients and 95% confidence intervals of the event-study version of the two-way-fixed-effects specification for employment separately for black, Hispanic, and white men.

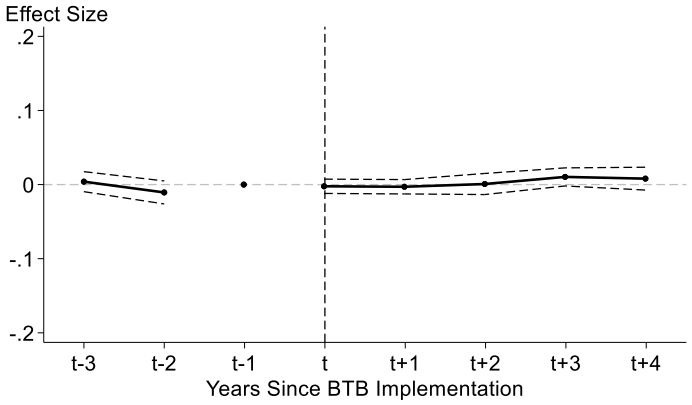


Figure 1: Black Men

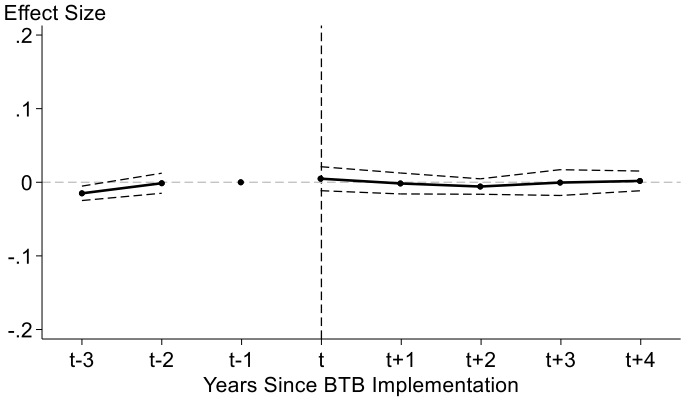


Figure 2: Hispanic Men

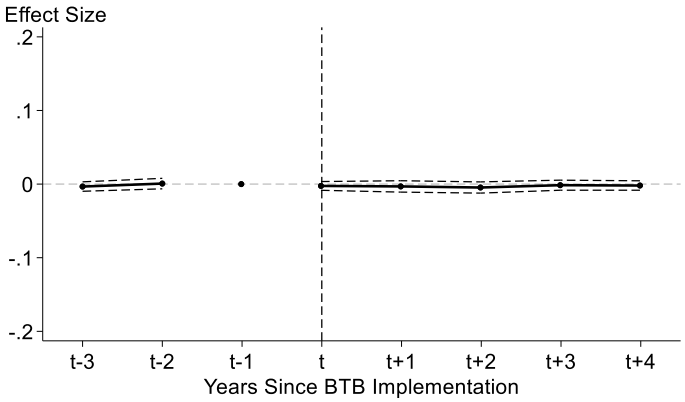


Figure 3: White Men

Neither black nor white men show any significant leads pre-treatment. However, Hispanic men show a significant negative t-3 lead (-0.0150, $p<0.01$), which vanishes in t-2.

To see whether the Hispanic t-3 lead reflects a genuine violation of parallel trends or is an artifact of the specification, I conducted a sequence of diagnostic checks (Appendix tables A.2, A.3 and A.4). I found that the result

was not caused by a singular state with a large Hispanic population (Appendix table A.2), nor was it an artifact of the MSA-specific linear trends included in the baseline specification (Appendix table A.3). Rather, the cause seems to be related to how congested the observations are.

Appendix table A.4. shows that 55.4% of the t-3 observations for Hispanic men came during the years 2008 and 2009, coinciding with the trough of the Great Recession. Re-estimating the event-study leads and joint test on a sample restricted to 2010 and later resolves the violation, as shown in Table 2 below.

Table 2: Pre-trend leads and joint F-statistic and p-value tests of Black, Hispanic and White men for the 2008-2019 and a shortened 2010-2019 sample.

	Black	Black (2010-2019)	Hispanic	Hispanic (2010-2019)	White	White (2010-2019)
t-3	0.0040 (0.0067)	0.0014 (0.0078)	-0.0150*** (0.0049)	-0.0054 (0.0066)	-0.0033 (0.0032)	-0.0038 (0.0039)
t-2	-0.0106 (0.0077)	-0.0044 (0.0092)	-0.0013 (0.0068)	0.0033 (0.0105)	0.0007 (0.0035)	-0.0022 (0.0041)
t-1	-0.0027 (0.0058)	0.0013 (0.0075)	0.0011 (0.0071)	0.0043 (0.0077)	-0.0062 (0.0038)	-0.0079* (0.0042)
N	181,511	155,630	189,065	163,383	765,829	640,991
Joint F-statistic	0.811	0.131	5.666	0.506	1.425	1.217
df	3, 50	3, 50	3, 50	3, 50	3, 50	3, 50
Joint p-value	0.4938	0.9411	0.0020	0.6797	0.2464	0.3133

*Standard errors in parentheses * p<0.10, ** p<0.05, *** p<0.01*

For Hispanic men, the t-3 coefficient shrinks from -0.0150 (s.e. 0.0049) to -0.0054 (s.e. 0.0066), and the joint p-value rises from 0.0020 to 0.6797 with the 2010-2019 sample restriction. For black men, the restriction moves the coefficients toward zero, with the p-value rising from 0.4938 to 0.9411. For white men, the joint test becomes marginally more favorable, with the p-value rising from 0.2464 to 0.3133. However, the t-1 lead, which was insignificant in the full sample (-0.0062, s.e. 0.0038), becomes marginally significant at the 10% level under the restriction (-0.0079, s.e. 0.0042).

Moving on to exogeneity and SUTVA, Doleac and Hansen (2020) tested whether pre-period characteristics predicted the likelihood or timing of BTB adoption and found no statistically significant relationships between local

labor market conditions and BTB implementation. Neither Burton and Wasser (2025) nor I revisit these tests. Both papers also examined moving patterns and found no evidence that BTB adoption is associated with migration for the target groups. I also confirm the migration concerns on my extended sample using the ACS mobility question. Table 3 shows the effect of BTB on moving, and none of the results are statistically significant.

Table 3: Effect of BTB on migration (2008-2019 ACS data)

	Any move	Interstate/abroad move
BTB x Black	0.0041 (0.0069)	0.0031 (0.0031)
BTB x Hispanic	-0.0005 (0.0043)	0.0001 (0.0039)
BTB x White	-0.0037 (0.0024)	0.0000 (0.0016)
N	1129985	884734
R^2	0.0312	0.0271

*Standard errors in parentheses * p<0.10, ** p<0.05, *** p<0.01*

With these results, I believe that there is sufficient evidence to validate the extended research design for black and white men, though I believe that some caution is warranted when interpreting the results for Hispanic men. To address the Hispanic t-3 lead, an additional specification has been added in the results which limits the sample to 2010-2019. In the next section I first replicate the ACS results of Burton and Wasser (2025) and then extend the analysis to also include the years 2015-2019.

5 Results

This section presents the empirical results in two stages. First, I replicate the ACS estimates from Burton and Wasser (2025) using their original 2004-2014 sample. Second, I re-estimate the same specifications on the extended 2004-2019 sample, incorporating the additional BTB adoptions through 2019 discussed in Chapter 3, and compare the resulting estimates to the original period.

Using the sample of low-educated men aged 25-34, I was able to replicate the ACS analysis of Burton and Wasser (2025). Regression estimates of Eq. (1) for their 2004-2014 sample are presented in Table 4 below. The results are divided into three race categories in black, Hispanic and white. Columns 2-5 include MSA linear trends, column 3 is limited to MSAs only, column 4 is limited to BTB-adopting MSAs only and column 5 limits the sample to years 2008-2014. The fifth column is the preferred specification of Burton and Wasser (2025), with the remaining columns being included for robustness.

Table 4: Replication of Burton and Wasser (2025) estimates: effect of BTB on the probability of employment (2004-2014 ACS data)

	(1)	(2)	(3)	(4)	(5)
BTB x Black	-0.0040 (0.0064)	-0.0061 (0.0087)	-0.0049 (0.0082)	-0.0043 (0.0085)	-0.0031 (0.0110)
BTB x Hispanic	0.0124 (0.0097)	0.0043 (0.0070)	-0.0039 (0.0080)	0.0000 (0.0070)	0.0129 (0.0079)
BTB x White	-0.0003 (0.0043)	-0.0040 (0.0040)	-0.0058 (0.0050)	-0.0057 (0.0048)	-0.0052 (0.0042)
N	937198	937198	619731	391914	648301
R^2	0.1532	0.1554	0.1400	0.1437	0.1620
Pre-BTB Mean: Black	0.5459	0.5459	0.5627	0.5459	0.5060
Pre-BTB Mean: Hispanic	0.7246	0.7246	0.7325	0.7246	0.6994
Pre-BTB Mean: White	0.7963	0.7963	0.7983	0.7963	0.7701
% Effect: Black	-0.73	-1.12	-0.87	-0.80	-0.61
% Effect: Hispanic	1.71	0.60	-0.54	0.00	1.85
% Effect: White	-0.04	-0.51	-0.73	-0.71	-0.68
MSA FE	X	X	X	X	X
Year-Region FE	X	X	X	X	X
Demographics	X	X	X	X	X
Fully interact with race	X	X	X	X	X
MSA linear trends		X	X	X	X
MSAs only			X		

BTB-adopting only				X	
2008 and later					X

*Standard errors in parentheses * p<0.10, ** p<0.05, *** p<0.01*

As noted in the literature review, none of the effects under any of the five specifications are statistically significant. Column 5 gives out a 0.31 percentage point (-0.61 %) decrease in the likelihood of employment for black men. For Hispanic men the effect was a 1.29 percentage point (1.85 %) increase and for white men a 0.52 percentage point (-0.68 %) decrease.

These estimates imply that BTB had very limited, if any, negative impacts on the employment of young low-educated black and Hispanic men. In fact, the employment likelihood increase for Hispanic men was the most significant finding at p = 0.1079, though it is worth noting that for alternate specifications in columns 2-4 the effect largely vanishes. While the short-term effects from ACS data might get attenuated due to the annual frequency, the results leave little doubt that any such effects during the sample period would be offset in the medium to long term.

I now turn to the extended 2004-2019 sample. Incorporating the additional years and BTB adoptions expands the full sample from 937,198 to 1,418,881 observations (column 1), giving the estimates considerably more statistical power. Due to my findings regarding the pre-trends of the Hispanic sample in Chapter 4, a sixth specification is added to the table, which limits the sample to 2010 and later. Since the full black and white sample passed the parallel trends test, I am treating column 5 as the preferred specification for white and black men, and column 6 as the preferred specification for Hispanic men. Table 5 presents the results.

Table 5: Effect of BTB on the probability of employment (2004-2019 ACS data)

	(1)	(2)	(3)	(4)	(5)	(6)
BTB x Black	0.0026 (0.0050)	0.0042 (0.0043)	0.0053 (0.0049)	0.0036 (0.0044)	0.0049 (0.0043)	0.0064 (0.0045)
BTB x Hispanic	0.0096** (0.0045)	0.0066** (0.0032)	0.0053 (0.0039)	0.0063* (0.0037)	0.0058 (0.0040)	-0.0004 (0.0061)
BTB x White	0.0041 (0.0030)	0.0023 (0.0025)	0.0000 (0.0030)	0.0021 (0.0025)	0.0028 (0.0021)	0.0034* (0.0019)
N	1418881	1418881	954129	1139387	1129985	953372
R^2	0.1512	0.1528	0.1357	0.1470	0.1571	0.1559

Pre-BTB Mean: Black	0.4953	0.4953	0.5491	0.4953	0.4541	0.4323
Pre-BTB Mean: Hispanic	0.7217	0.7217	0.7389	0.7217	0.7005	0.6848
Pre-BTB Mean: White	0.7778	0.7778	0.7897	0.7778	0.7541	0.7394
% Effect: Black	0.53	0.85	0.96	0.73	1.09	1.49
% Effect: Hispanic	1.33	0.92	0.72	0.88	0.82	-0.06
% Effect: White	0.52	0.30	0.00	0.27	0.37	0.46
MSA FE	X	X	X	X	X	X
Year-Region FE	X	X	X	X	X	X
Demographics	X	X	X	X	X	X
Fully interact with race	X	X	X	X	X	X
MSA linear trends		X	X	X	X	X
MSAs only			X			
BTB-adopting only				X		
2008 and later					X	
2010 and later						X

*Standard errors in parentheses * p<0.10, ** p<0.05, *** p<0.01*

When extending the cutoff date from 2014 to 2019, the results on average shift slightly upward for white and black men, while Hispanic men see a slight decline. Column 5 now gives out a 0.49 percentage point (1.09 %) increase in employment likelihood for black men and a 0.28 percentage point (0.37 %) increase for white men, while column 6 gives a 0.04 percentage point (-0.06%) decrease for Hispanic men.

While the effects remain mostly statistically insignificant, some regressions now yield a statistically significant result either at the 10% or 5% level. For Hispanic men the positive employment effects are significant at the 5% level for two and at the 10% level for one of the specifications and for white men at the 10% level under one specification.

It is worth noting that Tables 4 and 5 report a total of five and six specifications respectively, each interacting with three racial groups. This raises a multiple hypothesis testing concern, and some false positives are expected by chance alone.

This concern is partly mitigated by the fact that the specifications are heavily correlated as opposed to completely independent. Still, given the pre-trend findings of Hispanic men outlined in Chapter 4 and the fact that the results for white men are not robust on more than one specification, I believe

that they do not present sufficient evidence against the null to be considered significant.

Table 6 further breaks down the sample estimates from columns 5 and 6 of table 5 by cutoff year. For black and white men, the coefficients shift steadily from small negative values toward small positive ones as the sample is extended. For Hispanic men the pattern runs the other way.

Table 6: Columns 5 and 6 of Table 5 for cutoff dates 2014-2019.

	2008-2014	2008-2015	2008-2016	2008-2017	2008-2018	2008-2019
BTB x Black	-0.0031 (0.0110)	0.0002 (0.0088)	0.0022 (0.0061)	0.0016 (0.0052)	0.0027 (0.0046)	0.0049 (0.0043)
BTB x Hispanic	0.0129 (0.0079)	0.0128** (0.0060)	0.0099** (0.0043)	0.0092* (0.0052)	0.0064 (0.0046)	0.0058 (0.0040)
BTB x White	-0.0052 (0.0042)	-0.0037 (0.0043)	-0.0010 (0.0029)	0.0029 (0.0025)	0.0022 (0.0023)	0.0028 (0.0021)
N	648301	743157	838429	934921	1032875	1129985
R^2	0.1620	0.1603	0.1590	0.1579	0.1571	0.1571
	2010-2014	2010-2015	2010-2016	2010-2017	2010-2018	2010-2019
BTB x Black	-0.0019 (0.0138)	-0.0024 (0.0098)	-0.0006 (0.0065)	0.0009 (0.0057)	0.0034 (0.0051)	0.0064 (0.0045)
BTB x Hispanic	0.0063 (0.0111)	0.0059 (0.0068)	0.0011 (0.0071)	0.0028 (0.0073)	0.0007 (0.0063)	-0.0004 (0.0061)
BTB x White	-0.0065 (0.0044)	-0.0033 (0.0046)	0.0011 (0.0024)	0.0046** (0.0023)	0.0032* (0.0019)	0.0034* (0.0019)
N	471688	566544	661816	758308	856262	953372
R^2	0.1592	0.1579	0.1569	0.1562	0.1557	0.1559

*Standard errors in parentheses * p<0.10, ** p<0.05, *** p<0.01*

Taken together, the tables point to a slight shift in the estimated effects of BTB across sample periods. Incorporating later years to the estimates shifts the employment outcomes of black and white men upward, while outcomes for Hispanic men trend downward. Regardless, the extended ACS sample analysis does not significantly change the narrative set in Burton and Wasser (2025) and Kaestner and Wang (2024) of BTB having very limited effects in employment outcomes of young, lowly educated men.

6 Discussion

The conclusion from replicating and expanding the ACS analysis of Burton and Wasser (2025) is that the null findings largely stay consistent between sample periods. When the cutoff date was extended from 2014 to 2019, the employment outcomes of white and black men shifted slightly upwards, while the employment outcomes of Hispanic men shifted slightly downwards. In the end, the measured coefficients remained small and insignificant in the preferred specifications.

A useful benchmark for interpreting these null results is the statistical discrimination theory. If BTB worked by reshuffling which demographic groups get hired, the total amount of employment should remain roughly unchanged while the coefficients for different groups should move in opposite directions. My results only give weak support for this pattern, and I find no strong evidence of systematic reallocation of employment by race.

Overall, these results align relatively well with the CPS estimates of Kaestner and Wang (2024). Their study differs both in regression equation and treatment definition, so the results are not directly comparable, but still give a useful benchmark to compare the findings to. Table 7 summarizes the tables 2-4 of Kaestner and Wang (2024) with the results divided into 2004-2014, 2014-2019, and 2004-2019 sample periods. These columns are further divided into two with the left column including month fixed effects, and the right column including region-by-month fixed effects.

Table 7: Kaestner and Wang (2024) estimates: effect of BTB on employment (2004-2019 CPS data)

	2004-2014		2014-2019		2004-2019	
BTB x Black	-0.034 (0.019)	-0.038 (0.022)	0.025 (0.020)	0.020 (0.021)	-0.021 (0.013)	-0.023 (0.016)
BTB x Hispanic	-0.015 (0.015)	-0.015 (0.018)	-0.021 (0.022)	-0.019 (0.023)	-0.007 (0.010)	-0.010 (0.013)
BTB x White	-0.010 (0.010)	-0.007 (0.008)	-0.001 (0.007)	-0.001 (0.007)	0.002 (0.005)	0.003 (0.004)
MSA FE	X	X	X	X	X	X
Month FE	X		X		X	
Region-by-month FE		X		X		X

*Standard errors in parentheses * p<0.10, ** p<0.05, *** p<0.01*

Kaestner and Wang (2024) measured a coefficient of -0.021 to -0.023 for black men, whereas my 2004-2019 ACS estimate ranges from 0.003 to 0.005 across specifications. For Hispanic men, Kaestner and Wang (2024) measured an effect of -0.007 to -0.010 against my 0.005 to 0.010. Finally, for white men the estimate of Kaestner and Wang (2024) was 0.002 to 0.003 against my finding of 0.000 to 0.004. None of the effects found by Kaestner and Wang (2024) were statistically significant. Using my preferred specifications as opposed to the full 2004-2019 ACS sample brings the estimated effect of Hispanic men closer to Kaestner and Wang (2024) at 0.000, while for black and white men the estimate remains within the boundaries of the full 2004-2019 sample.

Comparing the 2014-2019 estimates of Kaestner and Wang (2024) to my Table 6 also shows mostly similar trends. Like my Table 6, the 2014-2019 coefficients of Kaestner and Wang (2024) for black men are upward sloping while the estimates of Hispanic men are downward sloping. For white men Kaestner and Wang's (2024) 2014-2019 estimate is essentially flat at -0.001, whereas my Table 6 measures a slight upward shift. Once again, the numbers are not directly comparable, but the fact that they are generally close to one another while following similar trends gives more credibility to the overall findings.

With my results, there are now three studies using two independent data sources that contradict the findings of Doleac and Hansen (2020). Kaestner and Wang (2024) showed that the negative CPS results do not hold up when the sample period is extended until 2019, Burton and Wasser (2025) showed that the results do not hold up when using ACS data for the 2004-2014 period and my results show that ACS data also fails to show any negative effects in the extended 2004-2019 period. Put together, these three studies present a much stronger claim against Doleac and Hansen (2020) than any one of them on their own.

There are some possible explanations for the limited effects. Most BTB laws only affect the public sector which causes them to have limited overall

effects since they only apply to a relatively small subset of the aggregate labor market.

On top of that, BTB remains difficult to enforce, which could result in noncompliance among the employers who have a strong aversion to people with criminal records. For instance, Schneider et al. (2022) and Herring and Smith (2022) found evidence of significant noncompliance among employers in Minnesota and California following their BTB laws. If this is a broad pattern, the expected effects of BTB laws would diminish as the compliers would mainly consist of employers who were always willing to hire people with criminal histories. This differentiates BTB from the removal of private signals like credit checks that employers could not access without directly asking the jobseeker.

Despite the findings of this thesis and the work of Burton and Wasser (2025) and Kaestner and Wang (2024) aligning, more research of BTB seems warranted. The ACS and CPS datasets used in all three studies rely on proxy-based demographic targeting and are not able to observe actual record holders. As such, while providing evidence against nationwide statistical discrimination, none of the studies can speak on how BTB affects people who actually hold records.

Regardless, the weight of evidence from the empirical findings of this thesis and the works from Burton and Wasser (2025) and Kaestner and Wang (2024) suggests that the original Doleac and Hansen (2020) findings were likely spurious rather than of a real effect. This is an especially relevant finding for policy debates that have used Doleac and Hansen (2020) as evidence on BTB backfiring for minority employment.

7 Conclusion

Ban the Box laws in the United States have seen exponential implementation across states and MSAs starting from the 1990s with the bulk of the legislation happening in the 2010s. The laws are intended to help ex-offenders find jobs but have also caused worries over potential statistical discrimination.

The prior research on the effects of BTB has found mixed evidence of statistical discrimination, with the strongest evidence found in very narrow samples like in Agan and Starr (2018). Out of the broader macro studies, Doleac and Hansen (2020) found evidence of statistical discrimination against young, lowly educated black and Hispanic men, which has been challenged by Kaestner and Wang (2024), who extended the CPS analysis through 2019, and Burton and Wasser (2025), who corrected BTB law coding for the original 2004-2014 ACS and CPS samples.

This thesis synthesized the works of Kaestner and Wang (2024), and Burton and Wasser (2025) by extending the BTB analysis of the bigger ACS dataset to years 2004-2019. My extended analysis failed to find evidence of statistical discrimination, which aligns with the results of Burton and Wasser (2025) and Kaestner and Wang (2024). Together, these findings build a compelling case that the findings of Doleac and Hansen (2020) were likely spurious and caused by the lack of statistical power of CPS.

These findings carry meaningful implications for the ongoing policy debate around BTB. Given that stable employment is linked to lower recidivism among ex-offenders (Berg and Huebner, 2011), and that recent studies find little evidence that BTB imposes economically meaningful costs on other demographic groups, the results are consistent with BTB being a low-risk policy tool. That said, like the studies it builds on, this thesis relies on demographic proxies rather than individual criminal records, so it cannot speak directly to how much BTB benefits the group it was designed to help. Closing that gap remains an important task for future research.

Artificial intelligence disclaimer & references

Use of generative artificial intelligence

Claude Sonnet 5 was used to automate some of the coding and iterate on the initial thesis draft. More concretely, I used it to map counties and cities to FIPS codes by giving it a list of counties and cities that implemented BTB laws between 2015-2019 that I sourced from Avery and Lu (2021) and a nationwide Metropolitan Statistical Area data file. I also used it to seek potential inaccuracies, typos and bad word choices on the thesis draft. I used its suggestions at my own discretion and artificial intelligence had no part in writing the initial draft, which still largely mirrors this submitted version.

References

Agan, Amanda, and Sonja Starr. "Ban the box, criminal records, and racial discrimination: A field experiment." *Quarterly Journal of Economics* 133, no. 1 (2018): 191-235.

Avery, Beth. "Research Supports Fair Chance Policies." National Employment Law Project (2024). Accessed June 26, 2026. https://www.nelp.org/app/uploads/2024/04/Research-Supports-Fair-Chance-Policies-Apr.2024.pdf.

Avery, Beth, and Han Lu. "Ban the box: U.S. cities, counties, and states adopt fair chance policies to advance employment opportunities for people with past convictions." National Employment Law Project (2021). Accessed June 26, 2026. https://www.nelp.org/app/uploads/2021/10/Ban-the-Box-Fair-Chance-State-and-Local-Guide-Oct-2021.pdf.

Bartik, Alexander, and Scott Nelson. "Deleting a signal: Evidence from pre-employment credit checks." *Review of Economics and Statistics* 107, no. 1 (2025): 152-171.

Berg, Mark T., and Beth M. Huebner. "Reentry and the ties that bind: An examination of social ties, employment, and recidivism." *Justice Quarterly* 28, no. 2 (2011): 382-410.

Burton, Anne, and David Wasser. "Revisiting the unintended consequences of Ban the Box." *Journal of Public Economics* 250 (2025): 105473.

Connerley, Mark, Richard Arvey, and Charles Bernardy. "Criminal background checks for prospective and current employees: Current practices among municipal agencies." *Public Personnel Management* 30, no. 2 (2001): 173-183.

Craigie, Terry-Ann. "Ban the box, convictions, and public employment." *Economic Inquiry* 58, no. 1 (2020): 425-445.

Doleac, Jennifer, and Benjamin Hansen. "The unintended consequences of 'Ban the Box': Statistical discrimination and employment outcomes when criminal histories are hidden." *Journal of Labor Economics* 38, no. 2 (2020): 321-374.

Harless, William. "'Ban the Box' Laws Make Criminal Pasts Off-Limits." *Wall Street Journal*, August 3, 2013. Accessed July 18, 2026. http://www.wsj.com/articles/SB10001424127887323997004578640623464096406.

Herring, Christopher, and Sandra Smith. "The limits of ban-the-box legislation." IRLE Policy Brief. Berkeley: Institute for Research on Labor and Employment, University of California, 2022.

Jackson, Osborne, and Bo Zhao. "The effect of changing employers' access to criminal histories on ex-offenders' labor market outcomes: Evidence from the 2010-2012 Massachusetts CORI Reform." Working Papers 16-30, Federal Reserve Bank of Boston, 2017.

Kaestner, Robert, and Xufei Wang. "Ban-the-box laws: Fair and effective?" *International Review of Law and Economics* 78 (2024): 106192.

Pager, Devah. "The mark of a criminal record." *American Journal of Sociology* 108, no. 5 (2003): 937-975.

Rose, Evan. "Does banning the box help ex-offenders get jobs? Evaluating the effects of a prominent example." *Journal of Labor Economics* 39, no. 1 (2021): 79-113.

Schneider, Lesley E., Mike Vuolo, Sarah E. Lageson, and Christopher Uggen. "Before and After Ban the Box: Who Complies with Anti-Discrimination Law?" Law & Social Inquiry 47, no. 3 (2022): 749-782.

Shoag, Daniel, and Stan Veuger. "Ban-the-box measures help high-crime neighborhoods." *Journal of Law and Economics* 64, no. 1 (2021): 85-105.

Uggen, Christopher, Mike Vuolo, Sarah Lageson, Ebony Ruhland, and Hilary K. Whitham. "The edge of stigma: An experimental audit of the effects of low-level criminal records on employment." *Criminology* 52, no. 4 (2014): 627-654.

Weiss, Deborah, William Dawson, Ronald McKinley, and Lee Webster. "Do ban-the-box policies increase the hiring of applicants with criminal records?" *PLoS ONE* 20, no. 4 (2025): e0320736.

Wozniak, Abigail. "Discrimination and the effects of drug testing on black employment." *Review of Economics and Statistics* 97, no. 3 (2015): 548-566.

Appendix

A. Table 1: Summary statistics of 25–34-year-old men by race and ethnicity and Ban-the-Box treatment status (2004-2019 ACS data)

	White		Black		Hispanic	
	(1) Control	(2) Treatment	(1) Control	(2) Treatment	(1) Control	(2) Treatment
Age	29.4702 (2.8890)	29.4493 (2.8859)	29.3677 (2.8795)	29.2929 (2.8827)	29.2569 (2.8959)	29.2068 (2.8795)
Enrolled in school	0.0704 (0.2559)	0.0797 (0.2708)	0.0802 (0.2716)	0.0921 (0.2891)	0.0823 (0.2748)	0.0944 (0.2924)
Less than high school	0.1394 (0.3464)	0.1226 (0.3280)	0.2639 (0.4408)	0.2169 (0.4122)	0.2427 (0.4287)	0.2304 (0.4211)
High school/GED	0.4743 (0.4993)	0.4677 (0.4990)	0.4514 (0.4976)	0.4492 (0.4974)	0.4320 (0.4954)	0.4231 (0.4920)
Live in an MSA	0.4099 (0.4918)	0.6853 (0.4644)	0.4777 (0.4995)	0.8216 (0.3829)	0.6645 (0.4722)	0.9066 (0.2911)
Northeast	0.0588 (0.2352)	0.1824 (0.3862)	0.0090 (0.0943)	0.1776 (0.3821)	0.0096 (0.0976)	0.1477 (0.3548)
Midwest	0.1246 (0.3303)	0.3160 (0.4649)	0.0214 (0.1447)	0.2143 (0.4104)	0.0233 (0.1507)	0.0965 (0.2953)
South	0.7140 (0.4519)	0.2836 (0.4507)	0.9624 (0.1902)	0.4754 (0.4994)	0.9223 (0.2677)	0.2175 (0.4125)
West	0.1026 (0.3034)	0.2180 (0.4129)	0.0072 (0.0847)	0.1327 (0.3392)	0.0448 (0.2069)	0.5383 (0.4985)
Employed	0.7711 (0.4201)	.	0.4657 (0.4988)	-	0.7164 (0.4508)	.
Employed (Pre BTB)	.	0.7788 (0.4150)	.	0.4953 (0.5000)	.	0.7217 (0.4482)
Employed (Post BTB)	.	0.7753 (0.4147)	.	0.5105 (0.4999)	.	0.7223 (0.4479)
N	181874	789286	49703	171320	39908	186791

Standard errors in parentheses

A. Table 2: Pre-trend coefficients of Hispanic men from Table 2 with the five states with most Hispanic men of the sample excluded one by one

State excluded	N	t-3	Joint F-statistic	df	Joint p-value
None (full sample)	189,065	-0.0150*** (0.0049)	5.666	3, 50	0.0020
CA	130,454	-0.0153** (0.0059)	2.480	3, 49	0.0720
TX	151,085	-0.0145** (0.0059)	6.757	3, 49	0.0007
FL	175,842	-0.0148** (0.0064)	3.514	3, 49	0.0219
NY	178,824	-0.0156*** (0.0049)	5.602	3, 49	0.0022
AZ	180,462	-0.0188*** (0.0050)	5.069	3, 49	0.0039

*Standard errors in parentheses * p<0.10, ** p<0.05, *** p<0.01*

A. Table 3: Table 2 estimates of Hispanic men with vs. without MSA-specific linear trends

	With MSA-trends	Without MSA-trends
t-3	-0.0150*** (0.0049)	-0.0179*** (0.0053)
t-2	-0.0013 (0.0068)	-0.0028 (0.0046)
t-1	0.0011 (0.0071)	-0.0011 (0.0053)
N	189,065	189,065
Joint F-statistic	5.666	5.109
df	3, 50	3, 50
Joint p-value	0.0020	0.0037

*Standard errors in parentheses * p<0.10, ** p<0.05, *** p<0.01*

A. Table 4: Number of Hispanic men in the t-3 cell by year, with mean employment in the t-3 cell versus the full Hispanic sample, 2008–2016

Year	N	Share	Cum. share	Mean employed (Hisp. t-3 cell)	Mean employed (Hisp. national)
2008	3,897	43.2%	43.2%	0.746	0.760
2009	1,096	12.2%	55.4%	0.701	0.709
2010	569	6.3%	61.7%	0.705	0.689
2011	173	1.9%	63.6%	0.682	0.651
2012	635	7.0%	70.6%	0.715	0.675
2013	1,462	16.2%	86.8%	0.726	0.691
2014	620	6.9%	93.7%	0.734	0.701
2015	452	5.0%	98.7%	0.750	0.716
2016	116	1.3%	100.0%	0.707	0.729